Evaluating Across-Hinge Dragging with Pen on Analog Curved and Foldable Devices

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Abstract
In this pictorial, we evaluate how different hinge radii impact dragging performance with a pen. A study examining a similar research question already exists. However, our main objective was to replicate the findings of that study using a lower budget while also focusing on movement variability as a parameter. To achieve this, an analogue prototype was built, using paper for the drawing surface. The movement variability was later calculated programmatically. It was found that the radius of the hinge has a significant impact on movement variability, even more than in the original study. We also examined how different dragging angles can affect movement variability.

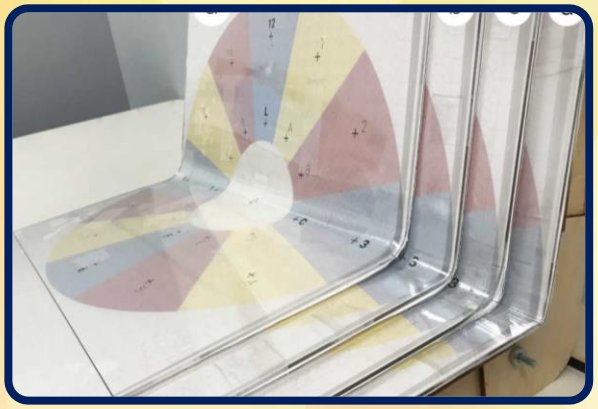


Figure 1.1: Prototypes (original study)

1 Motivation
Inspired by the seminal work titled "Evaluating Across-Hinge Dragging with Pen and Touch on Curved and Foldable Displays“ [1], this study seeks to replicate and extend the findings, regarding the influence of hinge radius on dragging performance. The original investigation explored the effects of dragging actions across different hinge radii (1mm, 6mm, 12mm, and 24mm) in a foldable 15-inch "laptop" prototype (see Figures 1.1 - 1.3).

The primary metrics examined include Drag Time, representing the duration required to complete dragging tasks, and Movement Variability, reflecting the degree of variation in movement. The study distinguishes between touch and pen input to evaluate dragging performance in varied usage scenarios.



Figure 1.2: Final study setup (original study)

Preliminary findings from the original research suggest, that the hinge radius plays a central role in shaping dragging performance, particularly with touch input. A larger hinge radius of 24mm demonstrates a notable reduction (13%) in Drag Time and decreased Movement Variability (7%) compared to smaller radii. Interestingly, while dragging along the hinge direction proves to be significantly faster (up to 30%) than dragging across, no significant effect of hinge radius on pen input performance is observed.

Introduction
Foldable displays represent a promising innovation in the realm of user interface design, offering enhanced flexibility and adaptability in device form factors. Understanding the impact of hinge radius on dragging performance is crucial for optimizing user interactions with foldable devices. Recent research has discovered the influence of hinge radius on drag performance in foldable displays and demonstrated its crucial role in shaping user interactions.



Figure 1.3: Building the prototypes (original study)

2 Basic idea and concept

Our evaluation revolves around replicating the findings of the paper while operating with a low budget. To achieve this, we plan to construct our analogue prototypes, replicating a low budget setup of the original study. However, our focus shifts from completion time to movement variability as the primary measure of performance assessment.

Research questions & hypotheses

Our research question centers on determining the optimal hinge radius for minimized movement variability in foldable displays: “What is the best performing hinge radius based on the movement variability?”. We hypothesize that as the hinge radius increases, movement variability decreases. Additionally, we anticipate observing higher movement variability in vertical drags over the hinge compared to horizontal drags. Moreover, we expect drags from the lower part of the apparatus to the upper part to exhibit more movement variability than the reverse direction. These hypotheses guide our investigation into the interplay between hinge radius and dragging dynamics, aiming to find key factors influencing user interactions with foldable displays.

Study design

The study design includes four conditions, each featuring varying hinge radii of 6mm, 12.5mm, 25mm, and 37.5mm. Quantitative measures primarily focus on assessing movement variability across these conditions, providing insights into the relationship between hinge radius and dragging performance. Additionally, qualitative measures include a post-study questionnaire aimed at capturing subjective user experiences and perceptions. The standardized questionnaire, NASA-TLX [2], serves as a reliable tool for evaluating the cognitive workload associated with interacting with foldable displays across different hinge radius settings. Through the combination of quantitative and qualitative assessments, our study tries to explore the impact of hinge radius on user interactions and usability in foldable display environments.

3 Study procedure

The study procedure begins with a warm welcome and introduction to the participants. They receive pre-study instructions to familiarize themselves with the tasks ahead. The study unfolds across all four conditions, with participants engaging in the main task specific to each condition.

Following completion of each condition, participants fill out a questionnaire, including the NASA-TLX [2] assessment, to assess their experience and cognitive workload. Upon completing all conditions, participants provide feedback through a post-study questionnaire.

Main task for participants

The main task for the participants was defined as follows: "Drag an imaginary folder from one point to another by drawing a line with a pen, as fast and precise as possible". The task includes connecting colored points, specifically from point 1 to point 2 (see Figure 3.1). Each participant is required to draw nine lines per paper, with three papers assigned per condition (see Figure 3.2). Participants are allowed only one attempt per line and trial, highlighting the importance of accuracy and efficiency in their drawing movements.

Counterbalancing

Counterbalancing is implemented to ensure fair treatment across participants and to reduce the potential effects of order or sequence on the results (see Figure 3.3). Each participant is assigned a unique "Participant ID" and starts with the different Tasks at a designated "Start Point". These starting points are marked at the paper as "Bereich 1" to "Bereich 4". The tasks reflect the 4 different conditions (hinge sizes), but the sequence of these variations differs between participants to minimize biases.

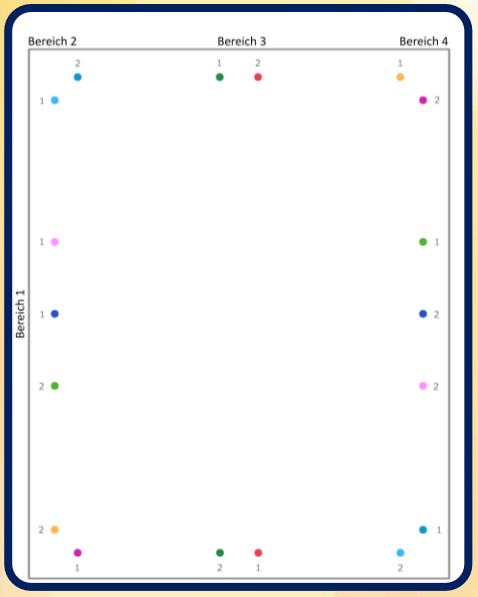


Figure 3.1: Uncompleted main task

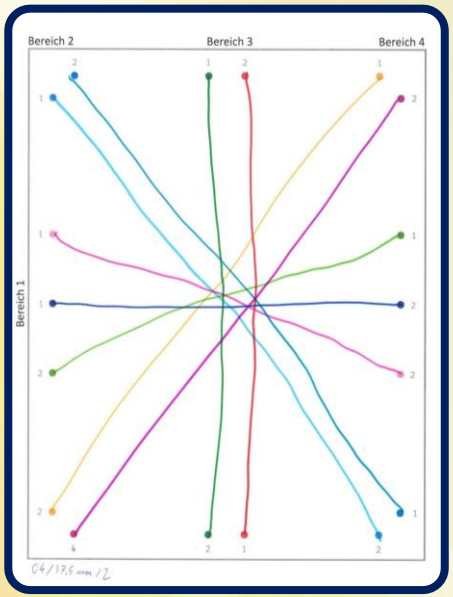


Figure 3.2: Completed main task

Paritcipant ID	Start Point	Task 1	Task 2	Task 3	Task 4
p1 p2 p3	Bereich 1	6 mm	12.5 mm	25 mm	37.5 mm
P4 p5 p6	Bereich 2	12.5 mm	25 mm	37.5 mm	6 mm
p7 p8 p9	Bereich 3	25 mm	37.5 mm	6 mm	12.5 mm
p10 p11 p12	Bereich 4	37.5 mm	6 mm	12.5 mm	25 mm

Figure 3.3: Overview counterbalancing

4 Building the prototype

Used Materials:

aluminium sheet - surface | wood - framing | pipes - bending the hinge | metal angles – hold wood together | glue - glueing materials together

Step 1 – Creating the surface

To create the surface of our prototype, we began by cutting the aluminium into suitable sized pieces (see Figure 4.1). We chose aluminium with a thickness of 3mm because of its flexibility, making bending easier and cost-effective. With dimensions of 1m by 60cm, we created four prototypes measuring 25cm by 30cm for each half of the screen. Instead of a cutter, we used a saw to avoid bending the metal to maintain its straightness. We then filed down the edges (see Figure 4.2) to prevent any risk of injury when using the prototype and utilized pipes to bend the aluminium, forming the desired hinges (see Figure 4.3). Pipes were selected because of their length, cheapness and thickness.

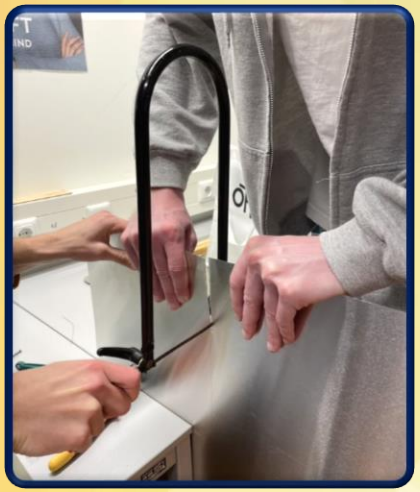


Figure 4.1: Cutting the aluminium



Figure 4.2: Filing the edges

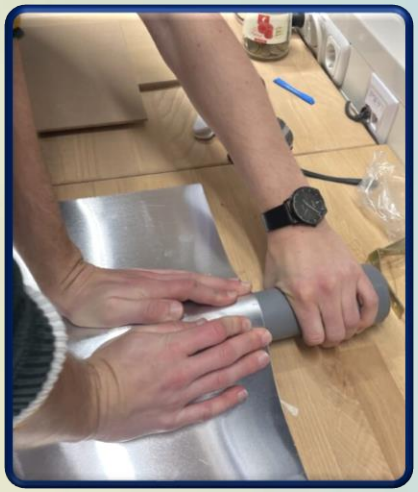


Figure 4.3: Bending the aluminium

Step 2 – Preparing the metal angles

We purchased metal pieces which initially required to be bend to a 105-degree angle by hand, mainly to use the same angle as the original study (see Figure 4.4). The initial idea was to experimented with a heater to make the bending of the metal easier, which proved ineffective. We picked metal angles due to their flexibility, affordability, and stability. We also roughened the surface so that the glue would adhere better to the metal angles later on (see Figure 4.5).



Figure 4.4: Bending the metal angles



Figure 4.5: Roughen the surface



Figure 4.6: Bent and roughened angles

Step 3 – Glueing everything together

Once all preparations were finished, including applying glue to both the metal angles (see Figure 4.7) and pre-cut wooden boards (see Figure 4.8), we proceeded to assemble them (see Figure 4.9). We chose a wooden back panel because of its durability, environmental friendliness, and resilience. We also selected a glue specialized for metal and wood for the glueing part, aiming for consistent and strong connections while also filling hollow spaces, ensuring convenience during prototype use.



Figure 4.7: Putting glue on metal angles

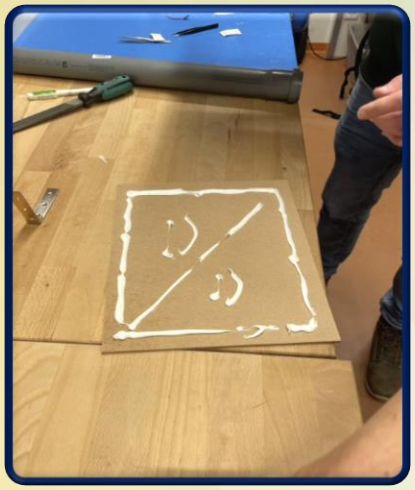


Figure 4.8: Glueing the wooden back panel

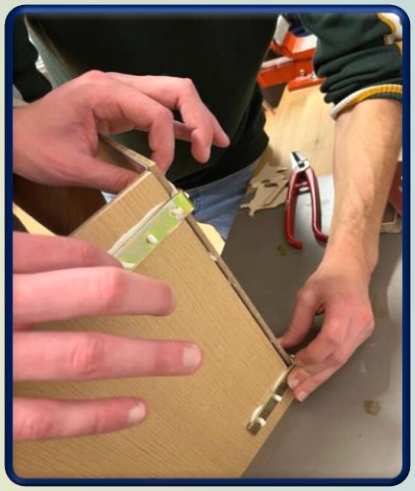


Figure 4.9: Putting everything together

Final Study Setup

After the glue dried overnight (see Figure 4.10), the final prototype was completed and ready for the evaluation (see Figure 4.11). With the help of clamps, we attached the papers to the prototypes, allowing us to switch the papers fast without risking holes in general. The paper, generously sponsored by the Campus Hagenberg, was both cost-effective and provided solid documentation, making it a preferred choice over electrical or video documentation, which would have required significantly more effort.

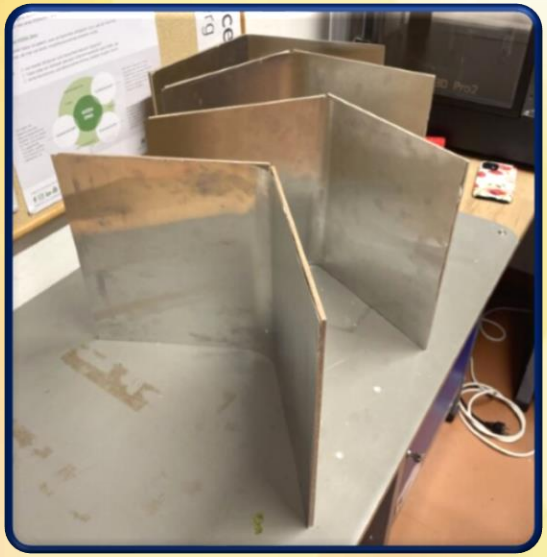


Figure 4.10: Letting the glue dry overnight

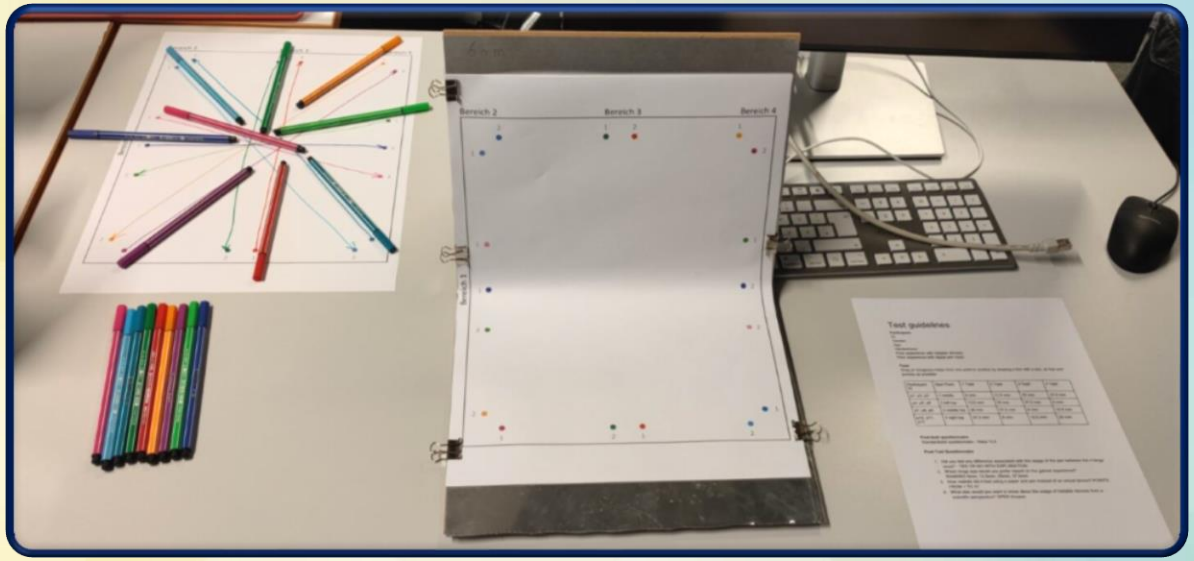


Figure 4.11: Final study setup

5 Evaluation

The evaluation of the produced qualitative data had to be split up into two parts: **already digital** (NASA-TLX, Post-Study-Questionnaire) and **analogue** (Test-Papers).

Analog to Digital (Paper to table)

Since the lines drawn by the participant are on paper, it first needs to be digitalized, to be then evaluated within a table.

Original (theoretical) approach:

Subtract every pixel on the paper that was there before the prototype test, from the paper after the prototype test, to extract the pixels drawn by the participant. The pixel changes per colour can then be divided by the according ideal line pixels, to find out, the difference in percentage.

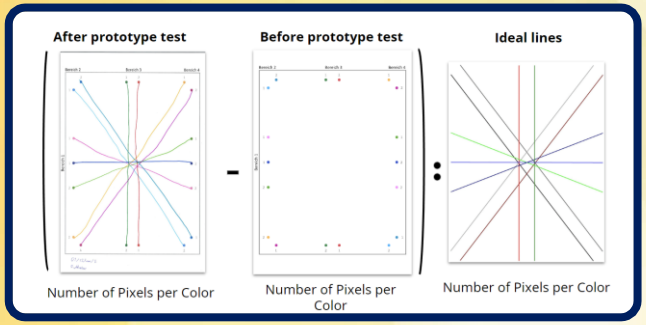


Figure 5.1: The original approach expected to calculate the movement variability correctly

Results using this method:

The original approach resulted in unexplainable and thus useless results when evaluated with the program.

The calculated movement variabilities lacked at least two important requirements on first sight:

1. When two lines are almost identical by eyesight, the movement variability should also be quite equal, since the only thing that is not consistent inside the formula, is the number of changed pixels by the attendant.
2. What is even more problematic is the fact that the movement variability can´t be negative by definition. Since the line can´t be more directly drawn than the ideal line, the movement variability can only be >= 0.

Color	MV 6mm	MV 12,5 mm	MV 25mm	MV 37,5mm
Task 2	-4,808041662	4,190383917	2,666828146	6,651326148
Task 3	16,66206516	33,92048592	39,69077858	51,93815572
Task 6	15,34933246	25,93441996	15,38755563	22,97485462
Task 7	-30,18977256	-19,52049833	-24,45072191	-23,49220146
Task 4	43,77996363	38,82046619	58,84857001	75,26037362
Task 5	18,66239625	38,05833936	45,05286359	66,13975273
Task 1	-27,6253905	-25,48319544	-25,42826736	-26,74310824
Task 9	-37,21807089	-28,65565229	-25,47473778	-27,85699596
Task 8	29,48176875	24,10554732	24,13976655	26,16250333

Figure 5.2: Impossible Results for attendant one
- The lines for Task 3 looked almost identical when reviewed, but showed big differences in the table
- All negative values are impossible by definition

These immense difficulties were not visible before, since everything was tested only using digital examples, except for one test with actual pen and paper, that did not point towards these problems.

Different usage of the original approach:

There were two other approaches, both using the original approach to get proper results. The first idea was to use the straightest line per colour, drawn by an attendant, for reference. The results were better, since it had fewer negative values than before, but there were still some present, so the whole table could not be considered as correctly evaluated.

The other try involved an algorithm to improve the colours by only using predefined ones. This, again, yielded better results, but the problem with the negative movement variability was still not solved.

Nevertheless, this was the start of the program, that was later used for the whole evaluation.

Difficulties finding a proper working method:

The overwhelming number of independent variables, which needed to be eradicated or controlled for a consistent evaluation, made this the most challenging task. The figures below show some examples for those problems, which eliminated any chance of an adequate usage of the original approach.



Figure 5.3: The colour differences inside a line are largely varying



Figure 5.4: Intersections can create other colours and falsify the result

Programmed Evaluation Method:

The various causes were identified, ranging from the easily fixable to the almost impossible, but it was imperative to find a consistently working solution to provide at least explainable results. The self-written refactored program consists of almost 700 lines of code, which were written in 8 days.

The functions, that created this example result, are explained on the following pages (F1 = Function 1).

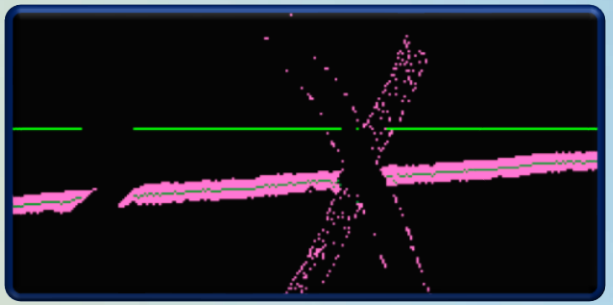


Figure 5.7: Example result for the colour rose, where light-green represents the ideal line and green the middle of the line drawn by the attendant. The difference in pixels between them is then representing the movement variability per x-coordinate

The program annihilates the following issues before providing the result:

1. *Colour differences - the whole line is completely reconstructed with the closest predefined colour*
2. *Gaps in line - even with big tolerance, the colour differences in the line were too high, so pixels belonging to the line were missed, causing gaps*
3. *Intersections – all intersections are completely ignored, so colour change problems are eliminated*
4. *Line can have gaps, ideal line doesn´t – if no middle line was found, no ideal line gets created*
5. *Line thickness – the thickness is now irrelevant, since both lines have a thickness of one pixel*



Figure 5.5: Ideal lines are drawn by a program, so they are perfectly consistent



Figure 5.6: Line thickness can vary due to pressure changes and/or fluidity of the pen

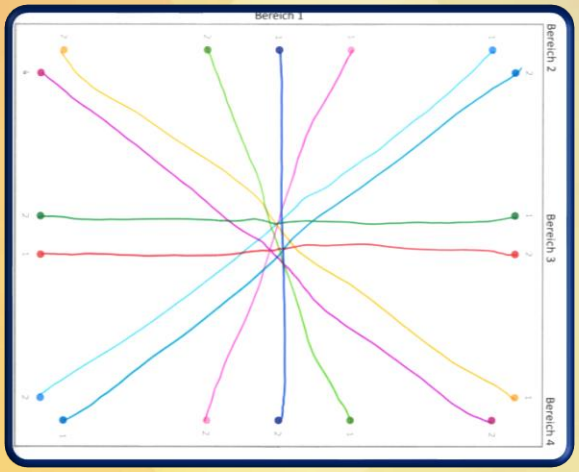


Figure 5.8: Scanning the paper with 300 dpi (JPG) and renaming it for use with the program (Folder structure: Attendant/Condition/Attempt)

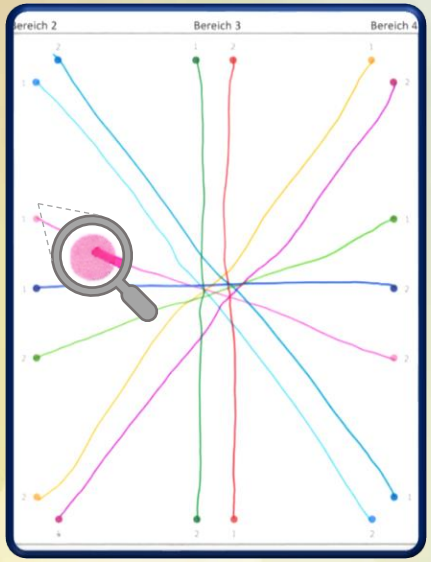


Figure 5.9: Rotating the image vertically (pixel by pixel), since the algorithm is written to work with vertical images



*Figure 5.10: Modifying **(F1,2)** the image only keeping the 128-grey value and finding the upper screen corners afterwards by highest similarity with predefined reference pictures. **(F3)** Knowing the upper left screen corner, the start and endpoints for the drag can be recalculated, eliminating all scanning offsets*

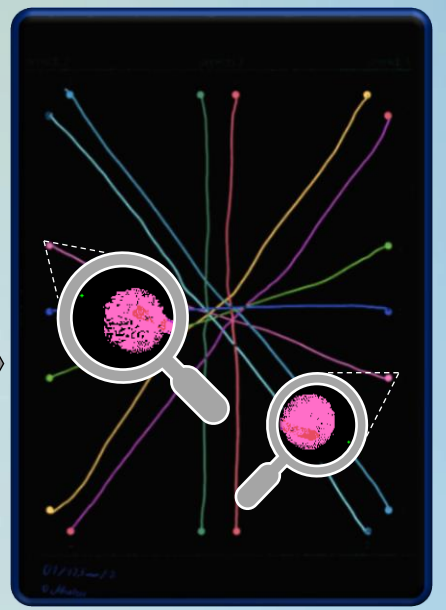


*Figure 5.11:. Afterwards the image gets modified, now containing only predefined colours **(F1)**. Now the angle between the start /endpoints can be calculated to place the reference points (green dots) outside the start/endpoints **(F4)***



*Figure 5.12: The image is now extended, to keep all details, and then turned around the angle. **(F5)** After the rotation, the reference points (green dots) are searched and their angle between each other is calculated once again, to eliminate any rotation errors.*

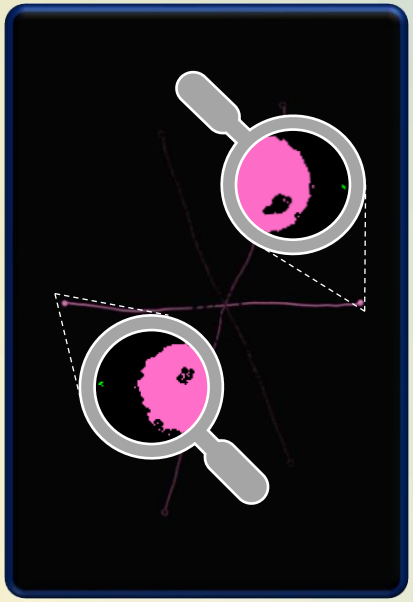


*Figure 5.13: Only one colour (rose) is kept in the next modification. **(F2)** Afterwards the reference points (green dots) need to be found once again, since they could have changed their position in the last rotation.*

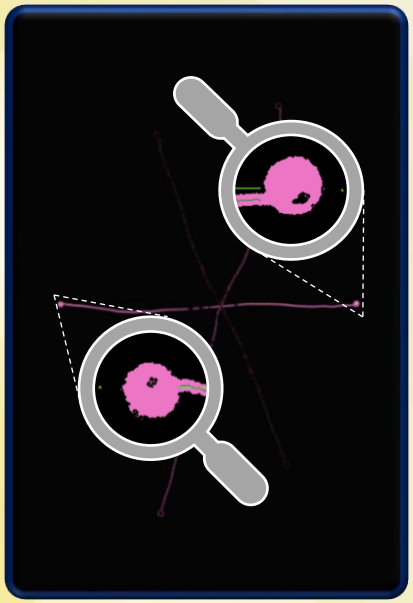


*Figure 5.14: Now the middle of the line (green) can be created, making sure that the start and endpoint, are not used within the algorithm, since they would cause problems. **(F6)***

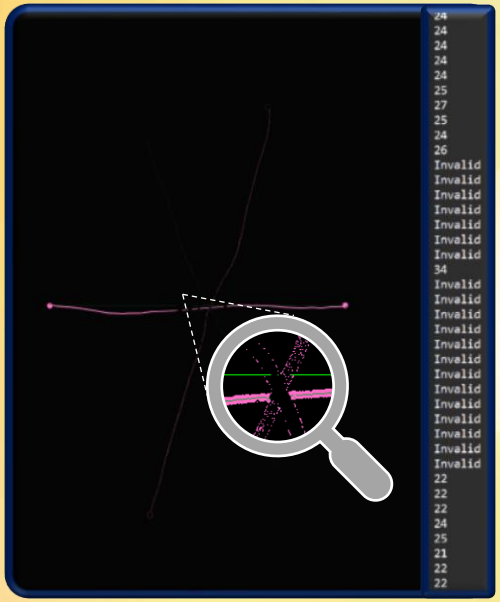


*Figure 5.15: The movement variability is now represented (for each x-coordinate in pixels), by the distance of the middle line (green) from the ideal line (light green). **(F6)** The movement variability is then saved in a text file, showing "Invalid" when no middle line was found. With this method, all before identified problems are now eliminated, so the movement variability can be brought into table format.*

Functions in detail:

Changing the image colours (F1):

Preprocessing: Before the prototyping test, the pen colours used in the test were predefined, using the pipette tool. The definition, however, needed to be adjusted after the user test. This became necessary due to the results the program provided. The final array of colours was then redefined by trial and error, keeping the best found values for each colour.
The tolerance is used for each R, G and B value separately and needs to be as high as possible, without interfering any predefined colour.

Function explanation:
The image gets iterated pixel by pixel, changing each pixel within a predefined tolerance (+/-55 for each R, G and B value) to a predefined colour.

Special requirement: If two predefined colours are inside the tolerance, the closest one is used. To choose the best suited one, the smallest difference of the vector length between the original colour and the predefined ones, got used. (colours are saved as 3-dimensional vector, so the vector length can be calculated by the formula: $sqrt(r^2 + g^2 + b^2))$

Result: The image now only contains predefined coloured pixels and black ones (black is used when no colour is in tolerance).

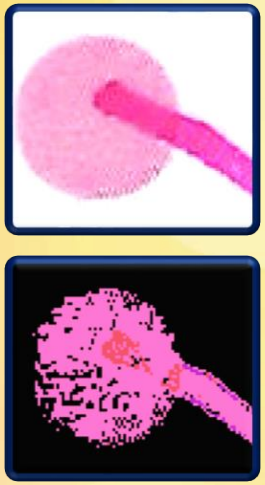


Figure 5.16: Example for a change of the colours for rose (which is noticeable quite similar to pink and red).

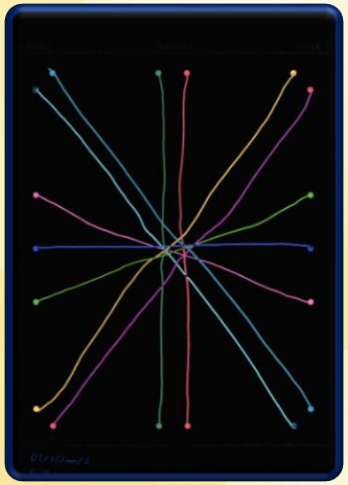


Figure 5.17: Image only containing predefined colours and black.

Modifying lines (F2):

Preprocessing: To get the best results from this function, it should be used after the image colours are changed **(F1)**. To ensure that the algorithm doesn't cause any expansion of the line (especially relevant for the outer parts), a minimum number of neighbours having the same colour needs to be declared.

Function explanation:
Another iteration over the whole image is necessary to sharpen the lines step by step. For each pixel, nine pixels (all eight around the currently checked one and itself) get checked. If five or more have the same colour, the currently checked pixel will be set to this colour, otherwise the colour stays the same as before. In the next iterations, the neighbours will then profit from the colour change, so step by step, gaps inside the line are closed. This is done until the abortion criterium is met, which takes ~8 iterations.

Special requirement: An abortion criterium (<10 changes in an iteration) needs to be defined, since an overoptimization can be very time consuming, without making much difference in the result.

Result: The lines in the image are now almost gap-free, so the following detection of the middle line **(F6)** is much more precise.

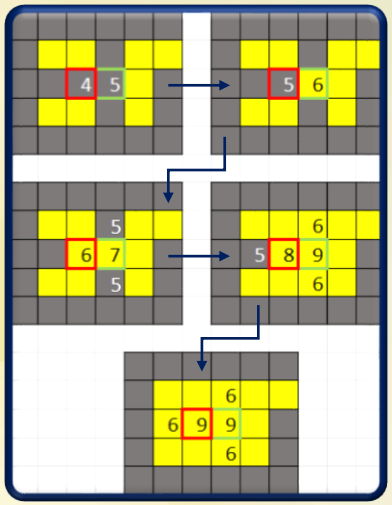


Figure 5.18: The improvement process of a line - the process happens from left to right (row by row), one iteration per image – so after 5 iterations there is no change anymore.

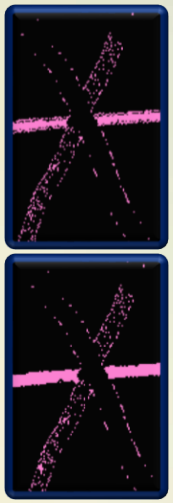


Figure 5.19: Example for a modification: before (top) and after (underneath).

Finding the screen reference points (F3):

Preprocessing: The reference points can be found via a difference check with predefined corners. To ensure a proper detection, this should be done after changing and modifying the colours **(F1,2)**, only keeping the grey values.

Function explanation:
For each pixel, all neighbours in the radius of 10 around the pixel are checked for equality with the reference images (21*21 px). If the colour at the mapped position is the same as the one at the referenced position, the counter gets incremented. Since this is done for each pixel in the original image, the highest similarity wins for each corner, so the reference points are always unique and correct.

Special requirement: Since the upper screen corners can only be found in the upper quarter, it is useful to use this value as an abortion criteria, to save time.

Result: The coordinates of those two corners are now known and can be used for offset calculations.

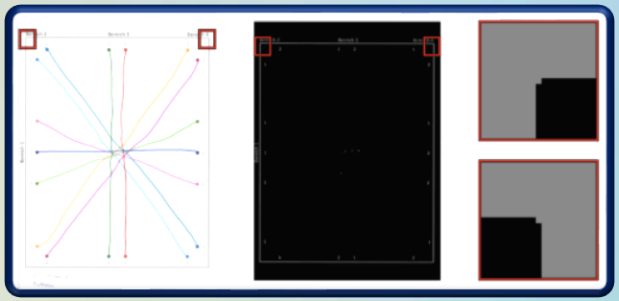


Figure 5.20: Scan offsets can be corrected by identifying the coordinates of the two upper screen corners – done by a similarity check with predefined reference images.

Placement of reference points for the rotation (F4):

Preprocessing: The distance for each start/endpoint is hardcoded. This is possible because the distance of the start/-endpoints from the upper left screen corner is always constant (since the paper was created in a vector program) when scan-offsets are eliminated.

Function explanation *(Figure on next page):*
A colour, which is guaranteed to not being present in the image (even with tolerance, is used for the cross representing the reference points. These are placed by programmatically enlarging the line over the start-/endpoint, each 100 pixels away from their location.

Result: The image now contains reference points needed for the correct rotation later.

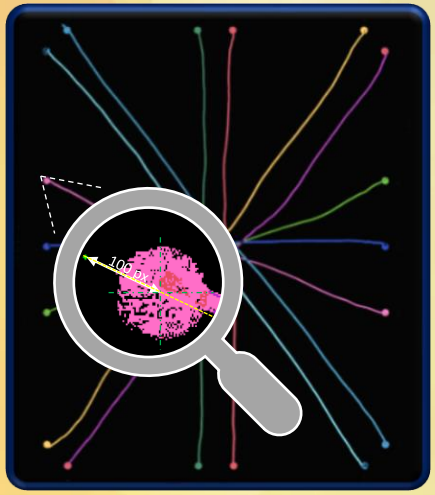


Figure 5.21: The reference points (green dots) are placed 100 pixels away from the centre (dark green lines) of the start/endpoint, in the programmatically calculated line (yellow).

Rotate the image around it´s centre (F5):
Preprocessing: The image needs to be rotated around it´s centre by the calculated angle between the according start and endpoint. Not only for the placement of the reference points, but also to ensure the correct rotation, the quadrant relation needs to be checked carefully. Since this is already done, while placing the reference points **(F4)**, any scanning offset is prevented beforehand, so the image is pixel perfect.
Function explanation:
To ensure that everything is visible correctly, the image first needs to be enlarged. To guarantee that it is big enough to show everything, the length of the diagonals are used as the new image size. Afterwards, it is inevitable to place the image that should be rotated, exactly in the centre of the new and bigger image. Then, the rotation can be started around it´s centre, using the pre-calculated angle, in order to maintain a pixel perfect rotation without missing details.
Special Requirement: The rotation doesn't need to be completely accurate by default, so it is required to rotate the image again, until the reference points (which can be found by searching for the predefined colour) are located exactly at the same height.
Result: A pixel perfect rotated and enlarged image containing the reference points.

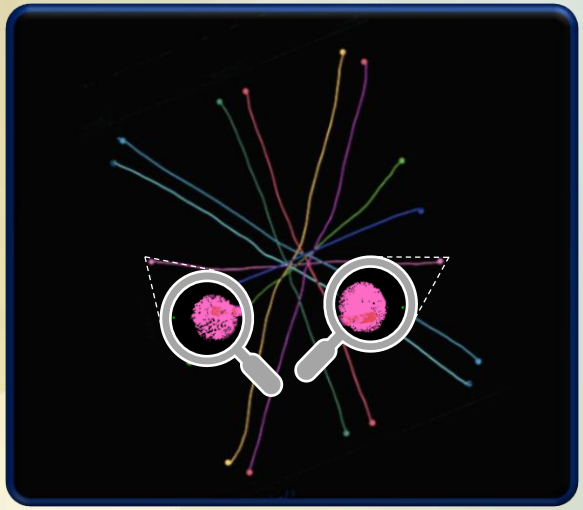


Figure 5.22: Pixel perfect rotated image for the colour rose. It also includes the reference points (green dots) to ensure the perfect result from the beginning

Calculate the movement variability (F6):
Preprocessing: The previous placed reference points **(F3)** can now be found using the according colour for the search. After the coordinates are known, the y-coordinate needs to be the same which is guaranteed **(F5). I**t is necessary to start not directly there, but to take an offset into account, so the start/endpoint itself (which could cause problems) is not in range anymore.

Function explanation:
To secure that the starting point is found correctly, the search algorithm starts with 100 pixels in each y-direction. It then remembers the last found pixel before exceeding the "gap-in-line-parameter" (maximal 3 black pixels, representing a gap in the line are allowed) which causes an abort. Afterwards, the mid of the two last remembered y-values gets calculated and the resulting pixel gets coloured (same colour as reference points) in the line. The movement variability is then the calculated mid pixel relative to the ideal line (line between the reference points).
Special Requirements:

- The movement variability can either be positive (underneath the ideal line), negative (above the ideal line) or 0 (exactly at the ideal line).
- To prevent mistakes in the detection, the allowed distance radius to search (10 when a line was found the x-coordinate before), gets incremented by one each time when no appropriate line is found. This results in a search shape like a hopper until the next valid middle point is found and the search radius is set back to the maximum thickness of the line estimated (10).

Result: The pixel perfect image now contains the middle line and the ideal line, eliminating almost all errors. The final image is then saved as a bitmap while the movement variability gets written in a file (for each x-coordinate in a separate line). If no middle line was found, the movement variability is "Invalid".

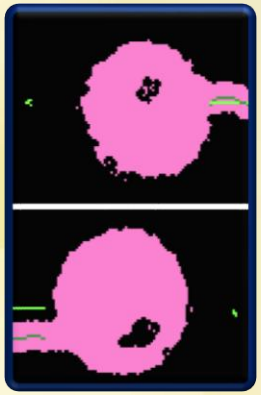


Figure 5.23:
The middle line gets only created after an offset from the reference points (130 pixels to be exact)

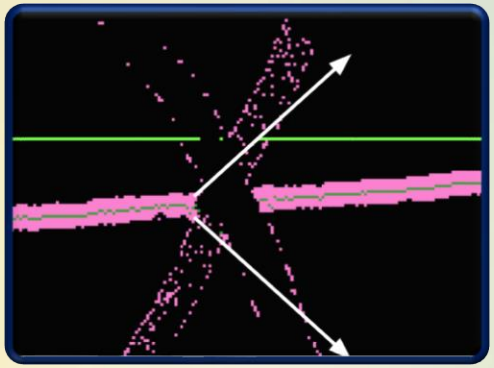


Figure 5.24:
Visible explanation of the hopper shape when no middle line is found. It also shows how little the inaccuracies are with this method (~5 pixels in this case)

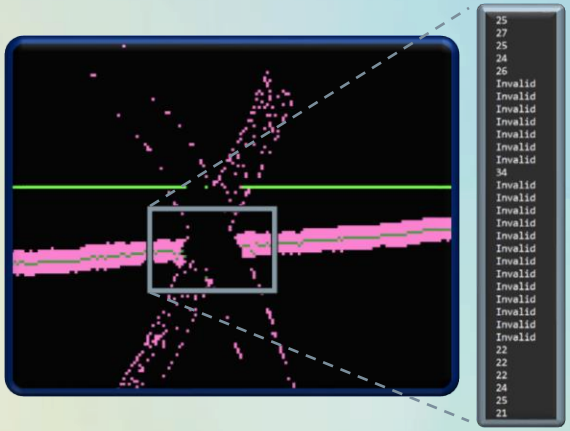


Figure 5.25:
Showcase of a bitmap for rose in the toughest part of the detection (an intersection with pink). The values representing the movement variability are saved in a .txt-file as presented in the grey framed image

	ID	6mm	12,5mm	25mm	37,5mm
1	1-1	39	54	49	57
2	1-2	47	37	53	37
3	1-3	40	56	58	55
4	2-1	54	77	45	46
5	2-2	56	67	57	28
6	2-3	59	63	41	40

Figure 5.26 : Extract of the table from jamovi [3], one data point represents the average movement variability per attempt

Descriptives

	6mm	12,5mm	25mm	37,5mm
N	36	36	35	36
Missing	0	0	1	0
Mean	56.1	59.1	60.1	52.6
Median	50.0	56.5	60	55.0
Standard deviation	22.3	17.4	17.8	17.8
Minimum	26	32	27	21
Maximum	106	97	95	90
Shapiro-Wilk W	0.910	0.952	0.981	0.975
Shapiro-Wilk p	0.006	0.118	0.795	0.566

Figure 5.27: Descriptive statistics of average movement variability per attempt

Analysis of Movement Variability

The statistical analysis of movement variability was done using the statistic tool *jamovi* [3]. In the study, 12 participants had to do 3 attempts for each condition (hinge radius).

Analysis with focus on hinge radius

For the comparison of different hinge radii, the average movement variability of all lines per attempt was calculated and stored in a table. An extract of the table is shown in Figure 5.26. First, the descriptive statistics were made. The results are displayed in Figure 5.27. Moreover, the Figure 5.28 shows the mean of movement variability for each hinge radius visually in a diagram.

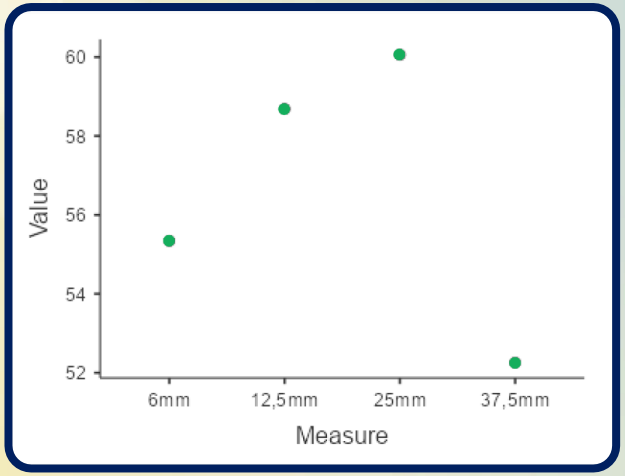


Figure 5.28: Mean of movement variability for each hinge radius

Subsequently, a Friedman test was carried out to see if there is statistical significance for the differences of movement variability between the 4 hinge sizes. The result of the Friedman test was $X^2 = 6.58$, df = 3, p-value = 0.087. Thus, the Friedman test did not show statistical significance. However, the ANOVA without the 6mm hinge condition showed statistical significance: $F(2,68) = 4.16$, p-value = 0.02.

Therefore, T-Tests were done with the 12.5mm, 25mm, and the 37.5mm hinge radii. The results of the T-Tests show that the movement variability of the prototype with the 37.5mm hinge radius is statistically significantly smaller than the prototypes with the 12.5mm and 25mm hinge radius. The Hypotheses and the results of the T-Tests are documented in the Table 1.

Hypothesis (text)	Hypothesis	Evaluation – T-Test
As the hinge radius increases, movement variability decreases.	12.5mm > 37.5mm	true, statistical significance t(35)=2.11, p<0.021
	25mm > 37.5mm	true, statistical significance t(34)=3.07, p=0.002

Table 1: Hypotheses and results of T-Test for analysis with focus on hinge radius

	ID	lightblue	turquoise	orange	pink	green	red	rose	lightGreen	darkBlue
1	1-1-6mm	34	32	64	79	26	30	20	43	23
2	1-1-12.5mm	150	62	102	20	37	20	50	36	5
3	1-1-25mm	102	96	39	48	43	21	39	49	5
4	1-1-37.5mm	124	116	23	50	61	24	35	74	9
5	1-2-6mm	47	49	60	76	26	32	67	50	16

Figure 5.29: Extract of the table from jamovi [3] , one data point represents the average movement variability per line per attempt

Analysis with focus on drag angles

Each line that the participant had to draw during the study has a different angle towards the hinge. For better differentiation, each line was drawn in a different colour. These colours are used to define the different conditions in this analysis.

For this analysis, the average movement variability of each line per attempt was calculated and stored in a table. The Figure 5.29 shows an extract from this. Next, the descriptive statistics of this data were analysed. One can see the results of the descriptive statistics in Figure 5.30. To better relate the results to the lines on the paper, the mean movement variability was written on the individual lines in Figure 5.31 (big image on the next page).

After the descriptive statistics, the data set was analysed using inferential statistics. The result of the Friedman test showed statistical significance (X^2=406, df = 8, p<0.001). Subsequently, Wilcoxon Signed-Rank Tests were performed to test the previously defined hypotheses. The results are shown in the Table 2. They are discussed in more detail in Section 6 *Summary & Discussion*.

Descriptives

	lightblue	turquoise	orange	pink	green	red	rose	lightGreen	darkBlue
N	144	141	143	143	144	144	143	143	143
Missing	0	3	1	1	0	0	1	1	1
Mean	74.3	76.9	77.7	74.9	38.4	35.9	58.1	56.5	18.2
Median	67.0	56	70	61	33.0	30.0	50	47	16
Standard deviation	42.0	58.0	45.9	47.7	21.4	23.0	31.3	36.3	11.3
Minimum	17	12	17	10	11	8	8	14	3
Maximum	212	247	241	271	171	171	163	196	69
Shapiro-Wilk W	0.931	0.836	0.909	0.927	0.825	0.838	0.911	0.842	0.849
Shapiro-Wilk p	< .001	< .001	< .001	< .001	< .001	< .001	< .001	< .001	< .001

Figure 5.30: Descriptive statistics of average movement variability per line

Hypothesis (text)	Hypothesis	Evaluation – Wilcoxon
Observing higher movement variability in vertical drags over the hinge compared to horizontal drags.	green_90° > darkBlue_0°	true, statistical significance p<0.001
	red_270° > darkBlue_0°	true, statistical significance p=0.001
Drags from the lower part of the apparatus to the upper part exhibit more movement variability than the reverse direction.	turquoise_225° > lightBlue_45°	True, no statistical significance, p=0.747
	pink_315° > orange_135°	False, no statistical significance, p=0.338
	red_270° > green_90°	False, no statistical significance, p=0.98

Table 2: Hypotheses and results of Wilcoxon Test for analysis of movement variability with focus on drag angles

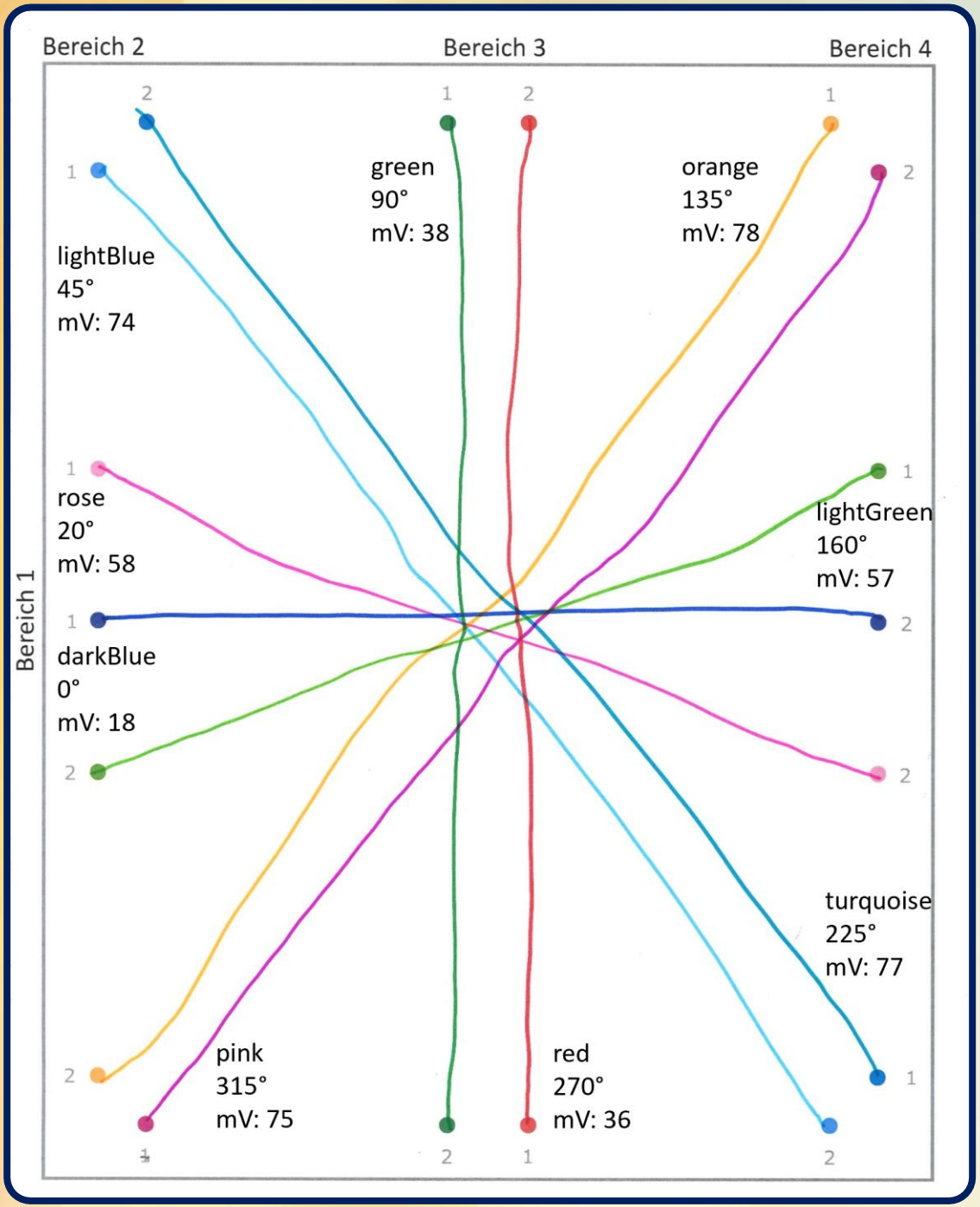


Figure 5.31: Mean movement variability per line

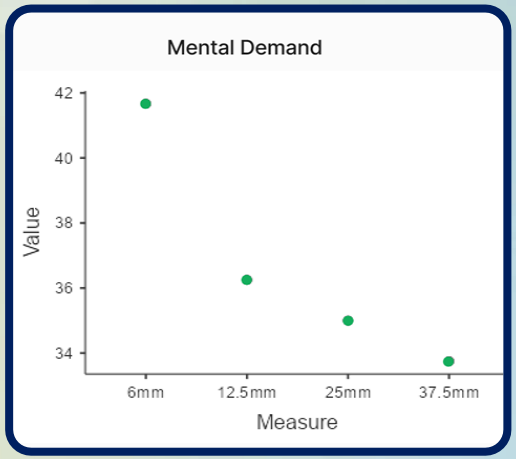


Figure 5.32: Average NASA TLX score for Mental demand

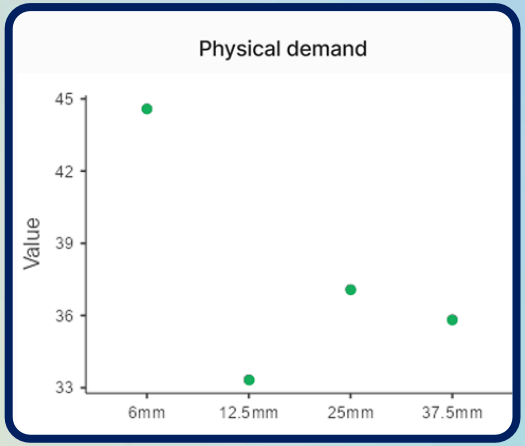


Figure 5.33: Average NASA TLX score for Phyical demand

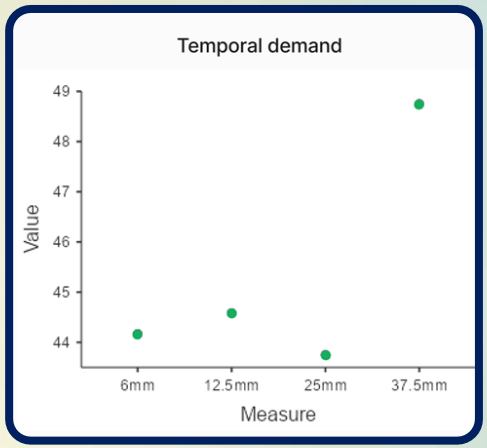


Figure 5.34: Average NASA TLX score for Temporal demand

NASA TLX Data

After each study iteration with one hinge size, the participant had to evaluate the subjective workload with the raw NASA TLX questionnaire [2]. The raw questionnaire contains the following six categories:

- Mental demand
- Physical demand
- Temporal demand
- Performance
- Effort
- Frustration

The participant selects a value for each category on a scale from low (0) to high (100) with increments of 5. To get a final NASA-TLX score, the average of the values of all categories is calculated.

In this study, the participant filled out the NASA-TLX on a self-built website (see Figures 5.38 and 5.39). By that, the data was already digital and could easily be copied into the statistics tool jamovi [3, in which all the descriptive statistics and inferential statistics were analysed.

For the descriptive statistics, one interesting result is that the mean and median of the NASA-TLX score get gradually smaller as the hinge radius increases (see Figure 5.40).

Since the Shapiro-Wilk Test indicated that the data is normally distributed, the ANOVA test was done to see if there is statistical significance in the differences between the hinge sizes. The result $F(3,33) = 1.37$, $p = 0.27$ (27%) showed that there is no statistical significance for the NASA-TLX score for hinge radius.

After analysing the average NASA-TLX score, the scores in the individual categories were evaluated (see Figure 5.32 - 5.37). To summarise, the score for the hinge radius of 37.5 mm has the smallest mean value in most cases. However, the inferential statistics for these scores also showed no statistical significance in any category.

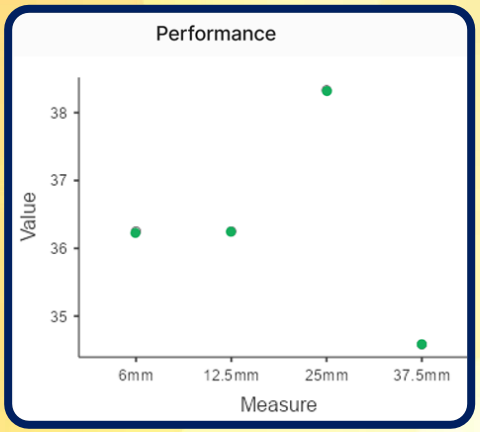


Figure 5.35: Average NASA-TLX score for Perfomance

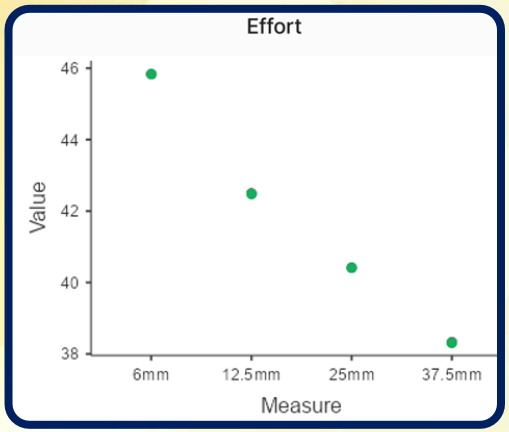


Figure 5.36: Average NASA-TLX score for Effort

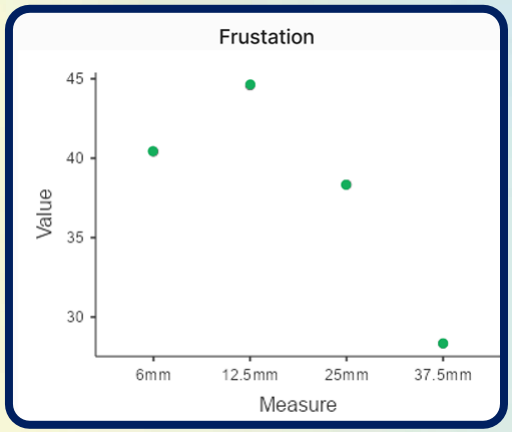


Figure 5.37: Average NASA-TLX score for Frustation

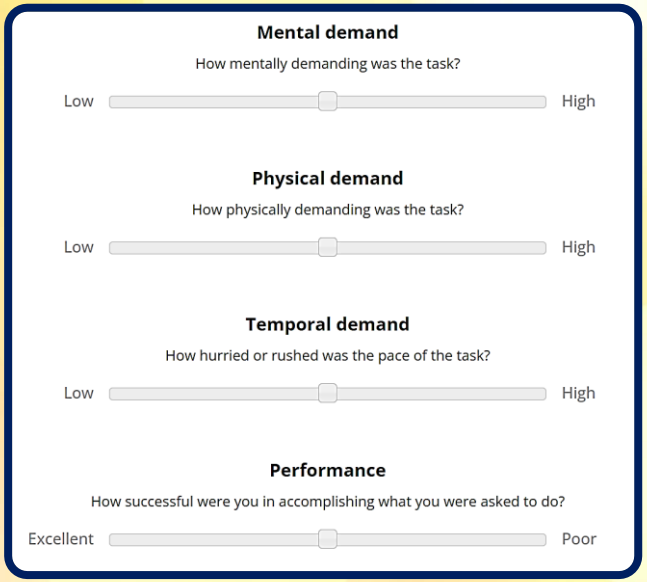


Figure 5.38: Extract of raw NASA-TLX

Step 3: resulting data

Proband: 01 **Task:** 25 mm

Demand	Rating	Weight	Product
Mental demand	5	1	5
Physical demand	60	1	60
Temporal demand	60	1	60
Performance	30	1	30
Effort	60	1	60
Frustration	35	1	35
		Product sum	250
		Total weights	6
		Rounded TLX score	**42**

Figure 5.39: Results of one NASA-TLX questionnaire

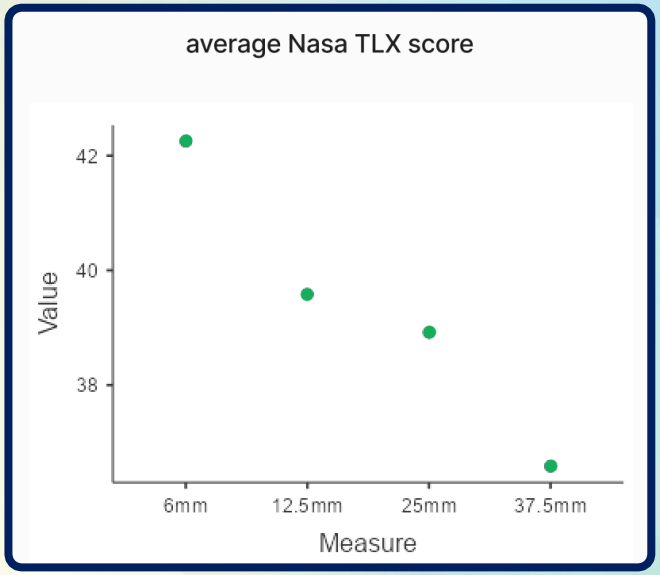


Figure 5.40: Average NASA-TLX score per hinge radius

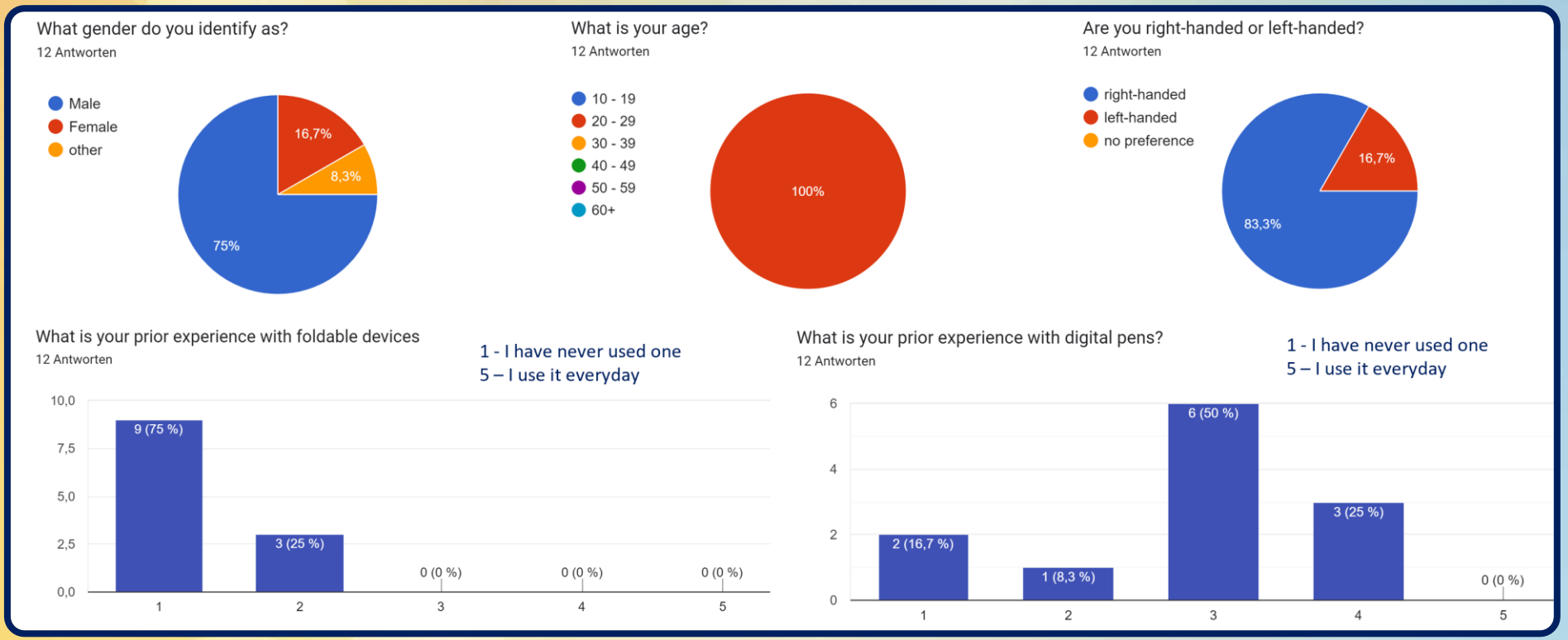


Figure 5.41: Results of demographic data and previous experience

Post-Study Questionnaire

After the study, the participants had to fill out a Google Form [4]. The goal of the questionnaire was to determine the subjective experience, perceptions, and opinions of the participants related to the previous study.

The questionnaire consisted of two parts. In the first part, the demographic data, gender, age and preferred hand were requested. In addition, questions about previous experiences with foldable devices and digital pens were asked. This data was taken to get an overview of the participants and to make the study comparable to potential future studies.

The second part was about the subjective personal study experience. First, the participants should say if they felt any difference between the 4 hinge sizes and, if yes, what the difference was. Then they should say, which hinge sizes they would prefer. These questions were asked regarding the research question. Moreover, one question was about their opinion about how realistic it felt using the prototype and a pen instead of an actual device. This question could show whether the results of this study are transferable to real devices. For further investigations, the last question was: *"What else would you like to know about the usage of foldable devices from a scientific perspective"?*

The evaluation of the data was also done with Google Forms [4]. Thus, the diagrams on this and the next pages are generated by it. The diagrams from the first part of the questionnaire are displayed in Figure 5.41. The demographic data is not evenly distributed. The participants were predominantly male and right-handed. In addition, they were all between 20 and 29 years old. The questions regarding previous experience with comparable devices show, on the one hand, that the participants have little experience with foldable devices and on the other hand, that they already have some experience with digital pens.

On the next two pages, one can find the results of the second part. 75% of the participants stated that they perceived a difference between the 4 hinge sizes (see Figure 5.42). Regarding the question: "*What was the difference between the different hinge sizes?*", 86% said that it was easier to draw on the devices with bigger hinge radii. P04: "*When it was more open, it was easier to draw a straight line.*" In Figure 5.44 one can see all the other statements regarding this question. Likewise, the results of the next question indicate that the majority of participants prefer the larger hinge radii (see Figure 5.43). About the question: "*How realistic did it feel using the prototype and a pen instead of a real device?*", the results can be divided into two groups. One said that it didn't feel realistic, while the other said that it came close to a real device (see Figure 5.45). The answers to the last question vary widely (see Figure 5.46). In summary, they show that foldable all-screen devices, as a new technology, offer potential for further research.

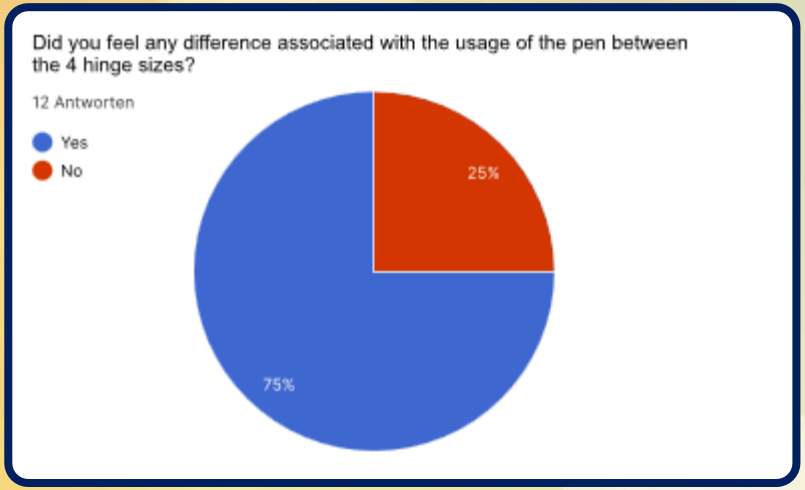


Figure 5.42: Results regarding whether a difference was perceived between the hinge sizes

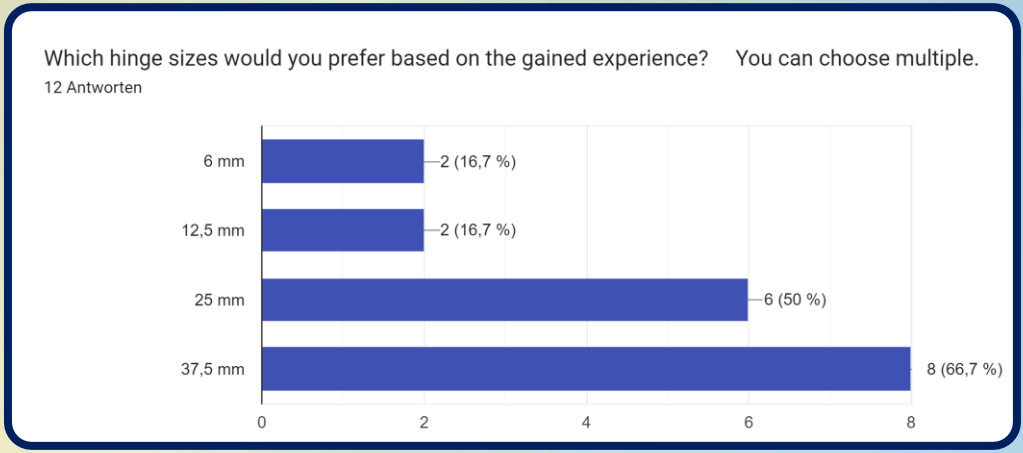


Figure 5.43: Results regarding which hinge sizes create a better experience

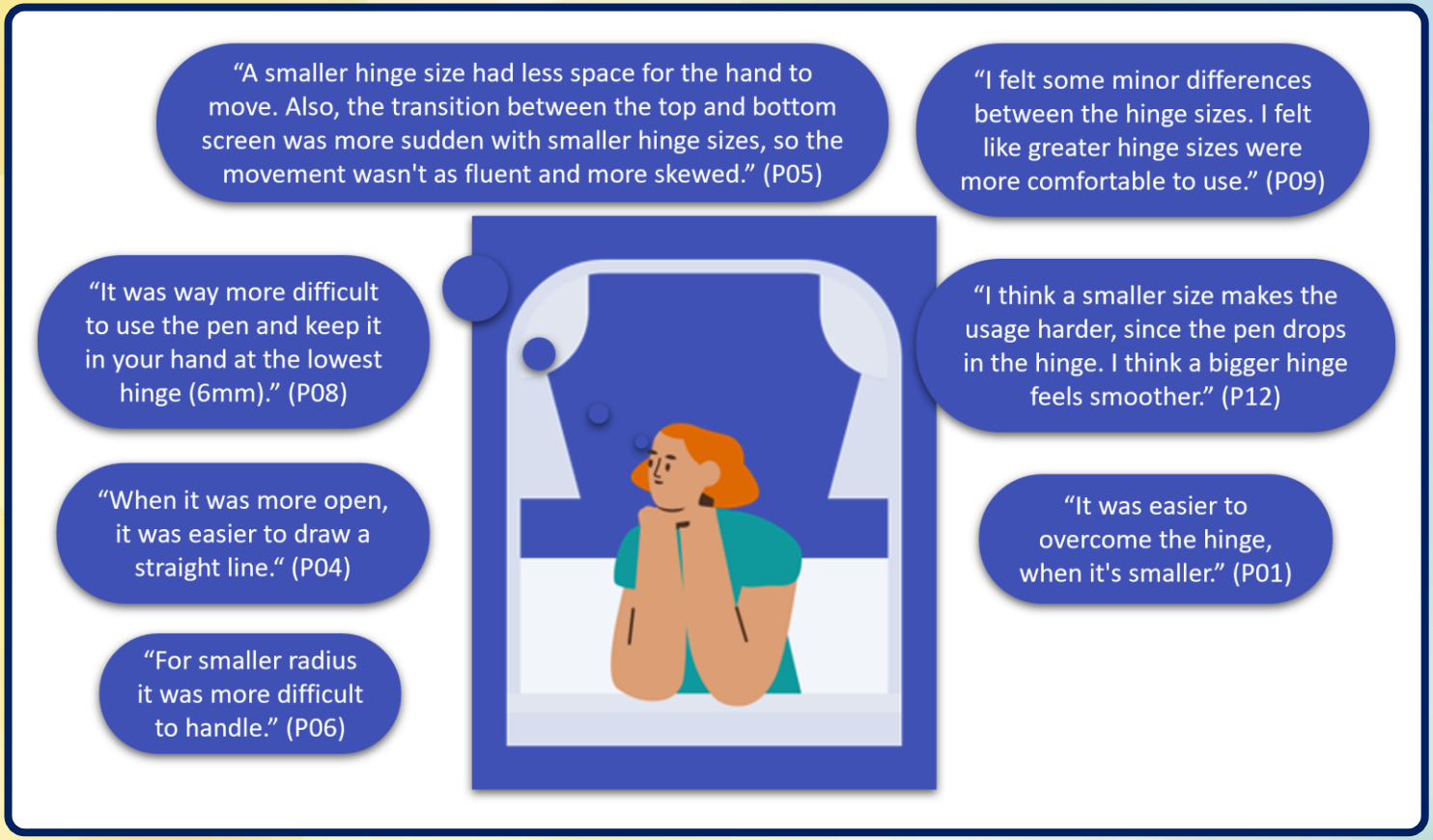


Figure 5.44: Answers of the participants to the question: "What is the difference between the 4 hinge sizes"?

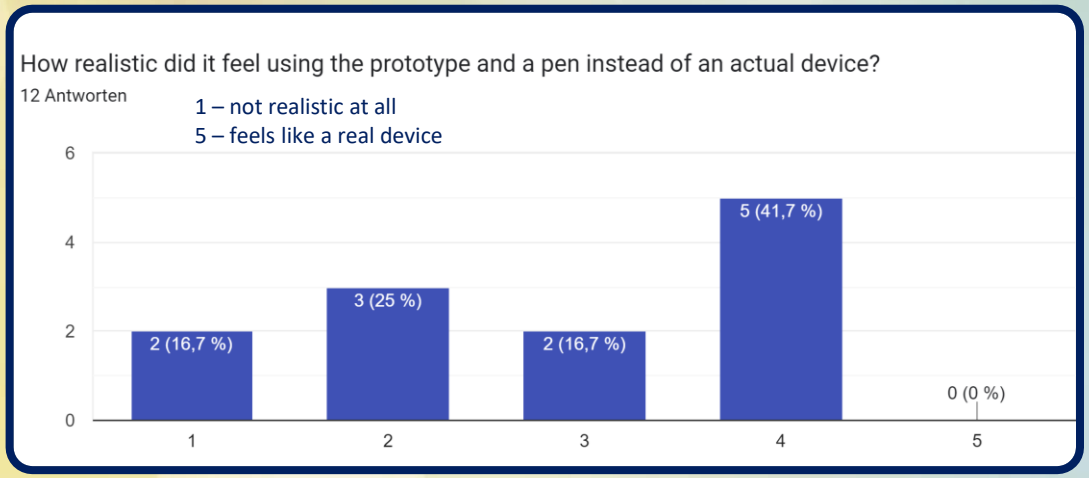


Figure 5.45: Results regarding the transferability of the prototypes to real devices

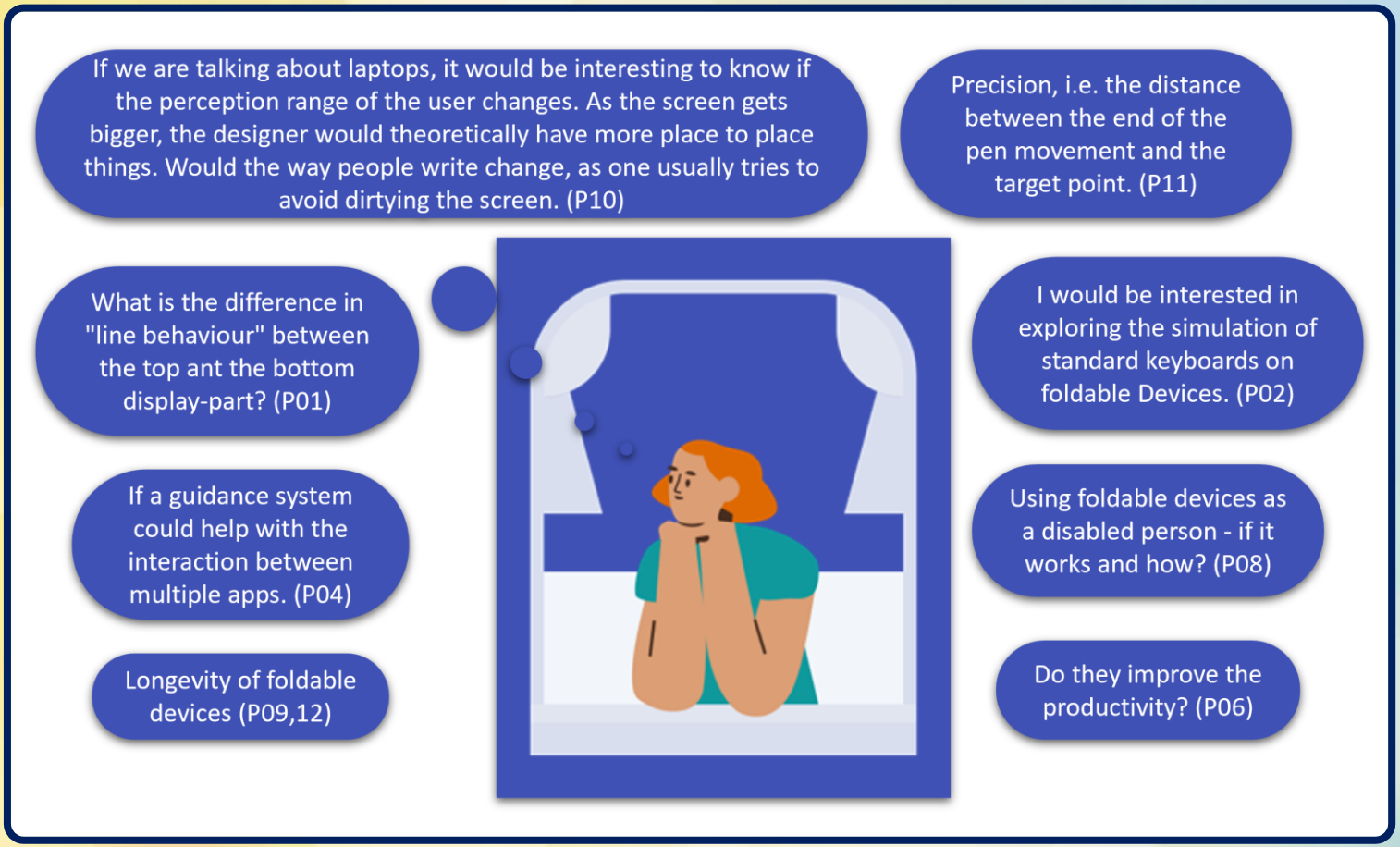


Figure 5.46: Answers of the participants to the question: What else would you like to know about the usage of foldable devices from a scientific perspective?

6 Summary & discussion

In this section, we will briefly discuss our findings regarding our prior hypotheses to validate or invalidate them.

Comparing hinge radii

As discussed previously, the device with the largest hinge radius (37.5mm) resulted in the lowest overall movement variability. However, it cannot be said that the movement variability decreases with the hinge radius. The prototypes with radii of 12.5mm and 25mm produce significantly higher movement variability than the one with a radius of 6mm, with 25mm having the highest variability overall.

One reason for this might be that the standard deviation for the 6mm prototype is significantly higher than those of the other three, being 22.3 compared to a range between 17.4 and 17.8. The maximum movement variability is also the highest for the 6mm hinge radius. This suggests that there might simply be more outliers in our data for the 6mm prototype. Since there was very limited testing data, the movement variability might smooth out to something that better fits our original hypothesis. However, this is also just speculation.

The results from both the questionnaire and the NASA-TLX differ from this phenomenon. For the NASA-TLX, the perceived workload decreases somewhat linearly the bigger the hinge radius gets.

The result is similar for the questionnaire. Here, just as many people preferred a hinge radius of 6mm as a radius of 12mm. However, significantly more people preferred the prototypes with larger hinge radii, with the radius of 37.5mm being the most liked one at 66.7%.

The answers to the question "*What is the difference between the 4 hinge sizes?*" also mostly supported this testament. P05, for instance, answered with "*A smaller hinge size had less space for the hand to move. Also, the transition between the top and bottom screen was more sudden with smaller hinge sizes, so the movement wasn't as fluent and more skewed.", while P12 said "I think a smaller size makes the usage harder, since the pen drops in the hinge. I think a bigger hinge feels smoother.”*.

In summary, our findings suggest that a hinge radius of 37.5mm is the most ideal one out of the 4 tested radii regarding movement variability. However, it has also to be noted that a larger hinge radius does not necessarily equal a lower movement variability, as we had hypothesised.

Comparing drag angles

When comparing the movement variabilities of different drag angles, it can be said that vertical drags across the hinge have a significantly higher movement variability than horizontal drags along the hinge. The two vertical lines resulted in movement variabilities that are 200% and 211% higher in comparison to the variability of the horizontal line. This also supports our prior hypothesis.

We had also hypothesized that drags from the lower part of the apparatus to the upper part have more movement variability than the other way around. This, however, turned out not to be true. Our setup had three lines that started at the bottom of the paper and ended at the top and three similar lines that started at the top and ended at the bottom. When comparing their movement variabilities, only one of the three lines that started at the bottom had a higher variability than its counterpart. All in all, the differences in movement variability regarding the start of the line were very small. Summing up, it can be said that whether the line starts on the top or the bottom of the screen has very little impact on the movement variability itself.

Comparing to previous study

All in all, our findings were a bit different than those of the referenced paper "*Evaluating Across-Hinge Dragging with Pen and Touch on Curved and Foldable Displays*" [1], but mostly the results were similar.

The prior paper was not able to find any significant difference in movement variabilities regarding the hinge radius when it came to pen input. They were only able to notice differences when using touch input. In our study, movement variabilities varied significantly across different hinge radii. The largest hinge size seems to be the most optimal one with our setup, which is also true for the prior study.

The prior study was also not able to find a reduction in movement variability for dragging horizontally along the hinge using pen input. In our case, we were indeed able to find significantly lower movement variabilities for horizontal drags compared to vertical ones.

Additionally, we also determined that whether the line starts at the top or the bottom of the screen has no significant impact on movement variability, which the previous study did not explore.

References

[1] Graeme Zinck u. a. „Evaluating Across-Hinge Dragging with Pen and Touch on Curved and Foldable Displays“. In: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. CHI ’23. Hamburg, Germany: Association for Computing Machinery, 2023. url: https://doi.org/10.1145/3544548.3580825

[2] Hart, S. G. & Staveland, L. E. (1988) Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In P. A. Hancock and N. Meshkati (Eds.) Human Mental Workload. Amsterdam: North Holland Press

[3] The jamovi project (2023). *jamovi* (Version 2.3) [Computer Software]. Retrieved from https://www.jamovi.org

[4] Google. (2023). Google Forms. [Software]. Google LLC. https://www.google.com/forms/about/